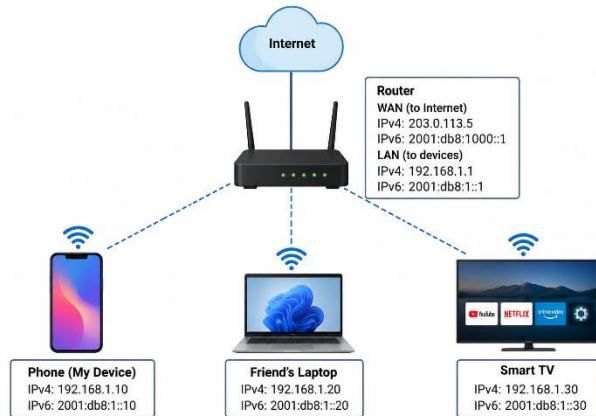


ip routing and routing protocols

1. Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.



2. Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.

Manual IP Routing Simulation

I simulated a network where my laptop and my friend's computer are in different subnets. The router checks its routing table and forwards the packet to the correct network.

Network Setup

- **My Laptop:** 192.168.1.10/24
- **Default Gateway:** 192.168.1.1
- **Router 1 LAN:** 192.168.1.1
- **Router 1 → Router 2:** 10.0.0.1
- **Router 2 → Router 1:** 10.0.0.2
- **Friend's Computer:** 192.168.2.10/24
- **Router 2 LAN:** 192.168.2.1

Routing Table – Router 1

Destination Network	Next Hop	Interface
192.168.1.0/24	Directly Connected	LAN
10.0.0.0/30	Directly Connected	R1-R2
192.168.2.0/24	10.0.0.2	R1-R2

Routing Table – Router 2

Destination Network	Next Hop	Interface
192.168.2.0/24	Directly Connected	LAN
10.0.0.0/30	Directly Connected	R2-R1
192.168.1.0/24	10.0.0.1	R2-R1

Step-by-Step Packet Travel

1. I send a packet from my laptop (192.168.1.10) to my friend's computer (192.168.2.10).
2. My laptop checks whether 192.168.2.10 belongs to its local network 192.168.1.0/24. It does not, so the packet is sent to the **default gateway 192.168.1.1**.
3. **Router 1** receives the packet and checks its routing table. It finds the entry: 192.168.2.0/24 → 10.0.0.2
4. Router 1 forwards the packet to **Router 2** through the 10.0.0.0/30 link.
5. **Router 2** receives the packet and checks its routing table. It sees that 192.168.2.0/24 is directly connected.
6. Router 2 forwards the packet to my friend's computer at **192.168.2.10**.
7. The friend's computer receives the packet and sends the response back through its default gateway, 192.168.2.1.

Packet Path

My Laptop

192.168.1.10

|

| 192.168.1.1

↓

Router 1

|

| 10.0.0.1 → 10.0.0.2

↓

Router 2

|

| 192.168.2.1

↓

Friend's Computer

192.168.2.10

Conclusion: The routers use their routing tables to determine the next hop. Since the destination 192.168.2.10 is in a different subnet, the packet is forwarded from my laptop to Router 1, then Router 2, and finally to my friend's computer.

2. Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

- Network Details

Branch	City	Network Address	Subnet Mask	Next Hop	Interface
Branch 1	Ahmedabad	192.168.10.0	255.255.255.0	Directly Connected	GigabitEthernet0/0
Branch 2	Mumbai	192.168.20.0	255.255.255.0	10.0.0.2	GigabitEthernet0/1
Branch 3	Delhi	192.168.30.0	255.255.255.0	10.0.0.3	GigabitEthernet0/2

Example Routing Table

Destination	Subnet Mask	Next Hop	Interface
192.168.10.0	255.255.255.0	Direct	G0/0
192.168.20.0	255.255.255.0	10.0.0.2	G0/1
192.168.30.0	255.255.255.0	10.0.0.3	G0/2

Explanation

The **Ahmedabad branch** uses the 192.168.10.0/24 network, the **Mumbai branch** uses 192.168.20.0/24, and the **Delhi branch** uses 192.168.30.0/24. When a packet needs to reach another branch, the router checks the destination network and forwards the packet to the appropriate **next-hop router** through the specified interface.

4. Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

Static Routing vs Dynamic Routing

Static routing means routes are manually configured by a network administrator, while dynamic routing protocols such as **RIP** and **OSPF** automatically learn and update routes.

Routing Type	Pros	Cons	Real-World Example
Static Routing	1. Simple for small networks. 2. Does not use routing protocol bandwidth.	1. Requires manual configuration. 2. Difficult to manage in large networks.	A small restaurant can use static routes to connect its billing computer, Wi-Fi router, and local server.

Routing Type	Pros	Cons	Real-World Example
	3. More predictable because routes are manually defined.	3. Does not automatically adjust when a route fails.	
Dynamic Routing (RIP/OSPF)	1. Automatically learns routes. 2. Can adapt when a network link fails. 3. Suitable for large and changing networks.	1. Requires more configuration and knowledge. 2. Uses CPU, memory, and network resources. 3. Routing updates can add network overhead.	A food-delivery service such as Zomato can have many servers and locations. Dynamic routing can help network traffic find available paths when a connection fails or the network changes.

Example

Suppose a food-delivery application has servers in **Ahmedabad, Mumbai, and Delhi**.

With **static routing**, the administrator manually specifies the route to each server. If the Mumbai network link fails, the route may need to be changed manually.

With **dynamic routing**, protocols such as **OSPF** can detect changes in the network and calculate another available route. This makes dynamic routing more suitable for large networks with many interconnected locations.

Conclusion: Static routing is useful for small and stable networks, while dynamic routing is generally more suitable for large networks where routes can change frequently.

5. Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

OSPF Routing Configuration Between Two Routers

AI-generated configuration using ChatGPT:

Router 1

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# interface g0/0
```

```
Router(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

```
Router(config)# interface g0/1
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# router ospf 1
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router(config-router)# end
```

Annotation

- enable → Enters privileged mode.
- configure terminal → Enters configuration mode.
- interface g0/0 → Selects the LAN interface.
- ip address 192.168.1.1 255.255.255.0 → Assigns an IP address to the interface.
- no shutdown → Turns on the interface.
- interface g0/1 → Selects the interface connected to Router 2.
- ip address 10.0.0.1 255.255.255.252 → Assigns the router-to-router IP address.
- router ospf 1 → Starts the OSPF routing process.
- network 192.168.1.0 0.0.0.255 area 0 → Advertises the LAN network through OSPF.
- network 10.0.0.0 0.0.0.3 area 0 → Advertises the connection between the routers.
- end → Exits configuration mode.

Router 2

```
Router> enable
Router# configure terminal
Router(config)# interface g0/0
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface g0/1
Router(config-if)# ip address 10.0.0.2 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit
```

```
Router(config)# router ospf 1
Router(config-router)# network 192.168.2.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router(config-router)# end
```

Annotation

- enable → Enters privileged mode.
- configure terminal → Enters configuration mode.
- interface g0/0 → Selects Router 2's LAN interface.
- ip address 192.168.2.1 255.255.255.0 → Assigns an IP address to the LAN interface.
- no shutdown → Activates the interface.
- interface g0/1 → Selects the interface connected to Router 1.
- ip address 10.0.0.2 255.255.255.252 → Assigns the router-to-router IP address.
- router ospf 1 → Starts OSPF.
- network 192.168.2.0 0.0.0.255 area 0 → Advertises Router 2's LAN network.
- network 10.0.0.0 0.0.0.3 area 0 → Advertises the link between the routers.
- end → Exits configuration mode.

Result: Both routers use OSPF to exchange routing information, allowing devices on the 192.168.1.0/24 network to communicate with devices on the 192.168.2.0/24 network.